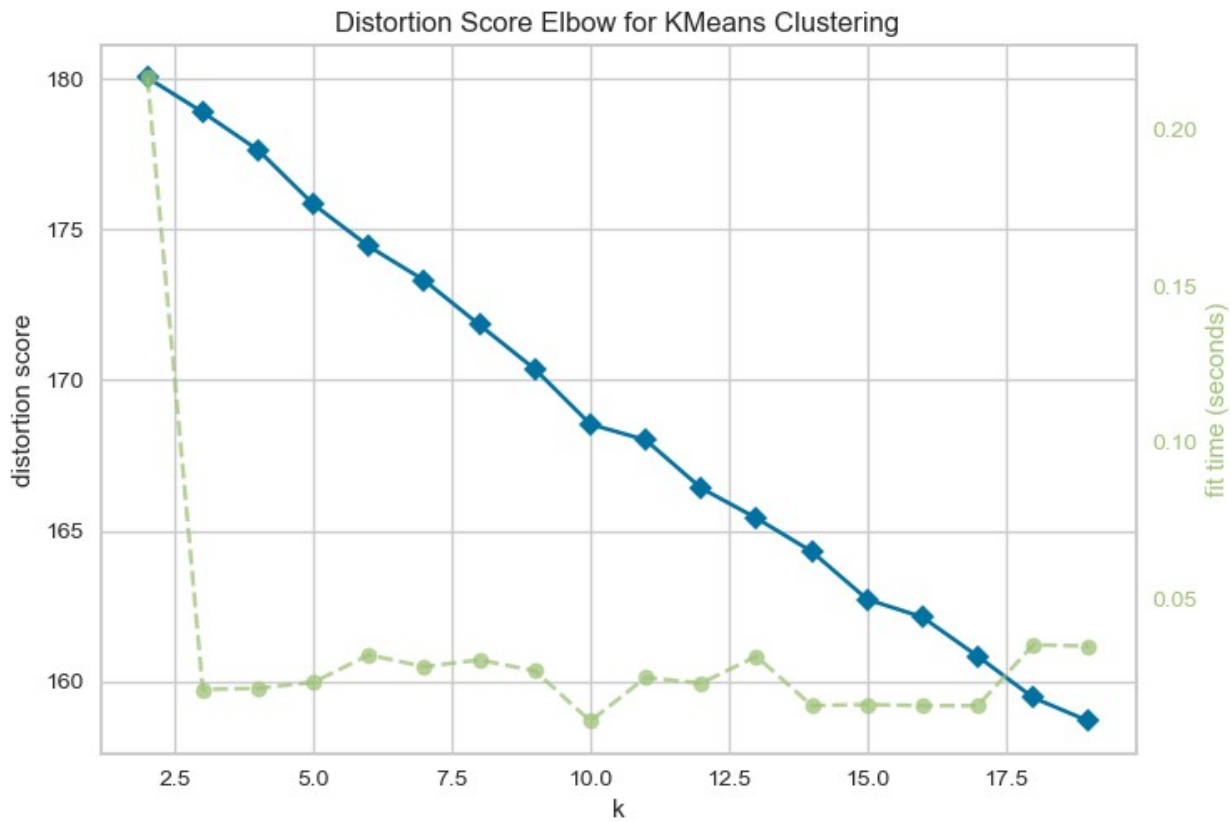
a. Appliquer l'algorithme de k-moyennes pour différentes valeurs de k

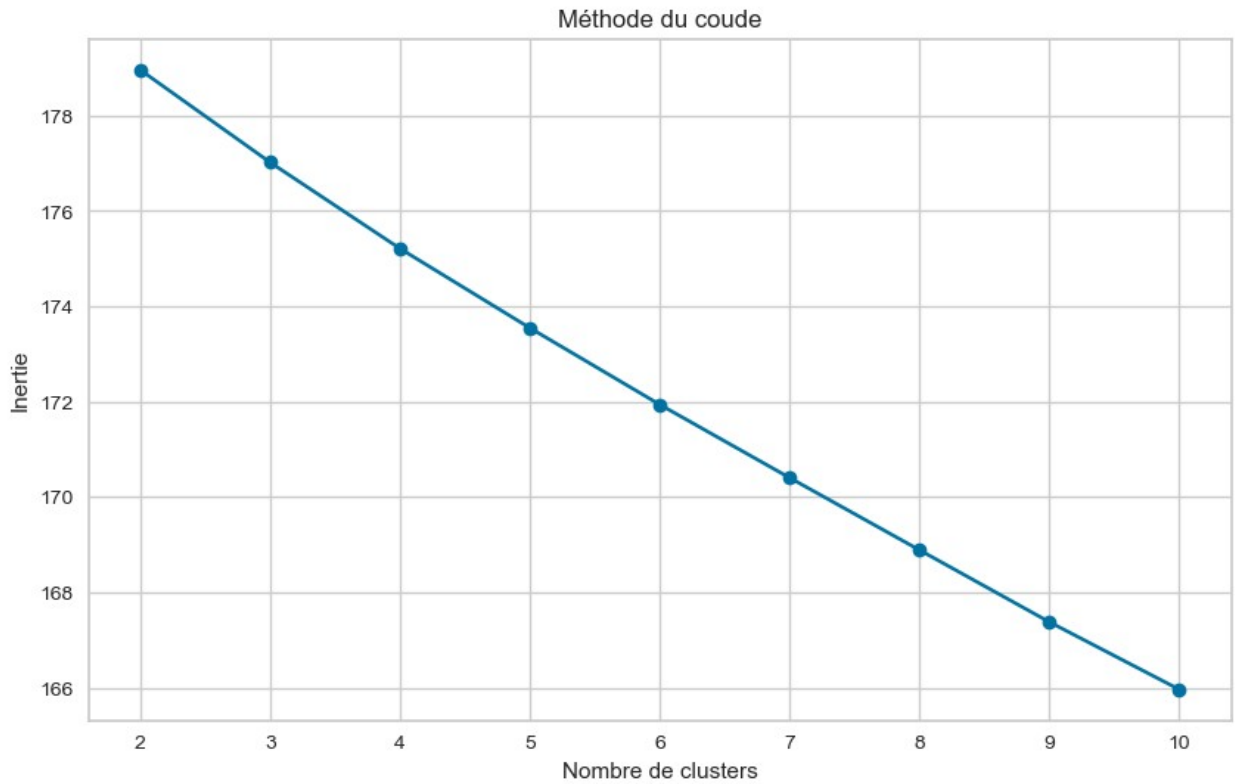
```
import warnings

warnings.filterwarnings('ignore')

import pandas as pd
import pickle
import numpy as np
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt
from yellowbrick.cluster import KElbowVisualizer, SilhouetteVisualizer

# Charger les données vectorisées
with open('./anthems_vectorized.pkl', 'rb') as f:
    X = pickle.load(f)

# Convertir la matrice sparse en une matrice dense
X_dense = X.toarray()

# Appliquer k-means pour différentes valeurs de k et visualiser avec
la méthode du coude
model = KMeans()
visualizer = KElbowVisualizer(model, k=(2, 20))

visualizer.fit(X_dense)        # Fit the data to the visualizer
visualizer.show()              # Finalize and render the figure

# Afficher le graphique de la méthode du coude
plt.show()
```



b. Déterminer le nombre optimal de clusters par la méthode du coude.

```
import matplotlib.pyplot as plt
from scipy.cluster.hierarchy import linkage, fcluster
from scipy.spatial.distance import pdist

# Calculer la matrice de linkage
Z = linkage(X_dense, method='ward')

# Calculer les inerties pour différentes valeurs de k
inertias = []
K = range(2, 11)

for k in K:
    # Créer les clusters
    labels = fcluster(Z, k, criterion='maxclust')

    # Calculer l'inertie
    inertia = 0
    for cluster in range(1, k+1):
        cluster_points = X_dense[labels == cluster]
        centroid = cluster_points.mean(axis=0)
        inertia += ((cluster_points - centroid) ** 2).sum()

    inertias.append(inertia)
```

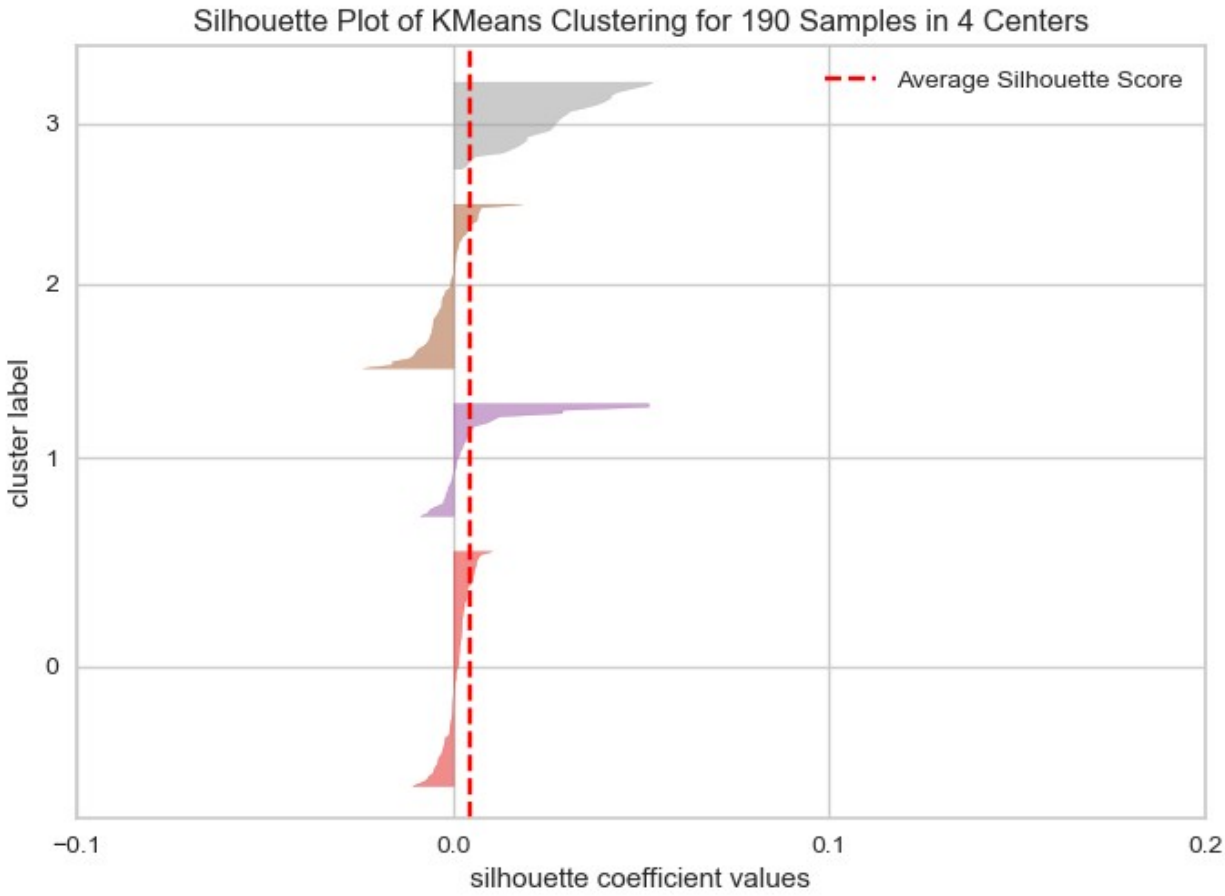
```python
# Tracer la courbe de la méthode du coude
plt.figure(figsize=(10, 6))
plt.plot(K, inertias, marker='o')
plt.title('Méthode du coude')
plt.xlabel('Nombre de clusters')
plt.ylabel('Inertie')
plt.show()
```

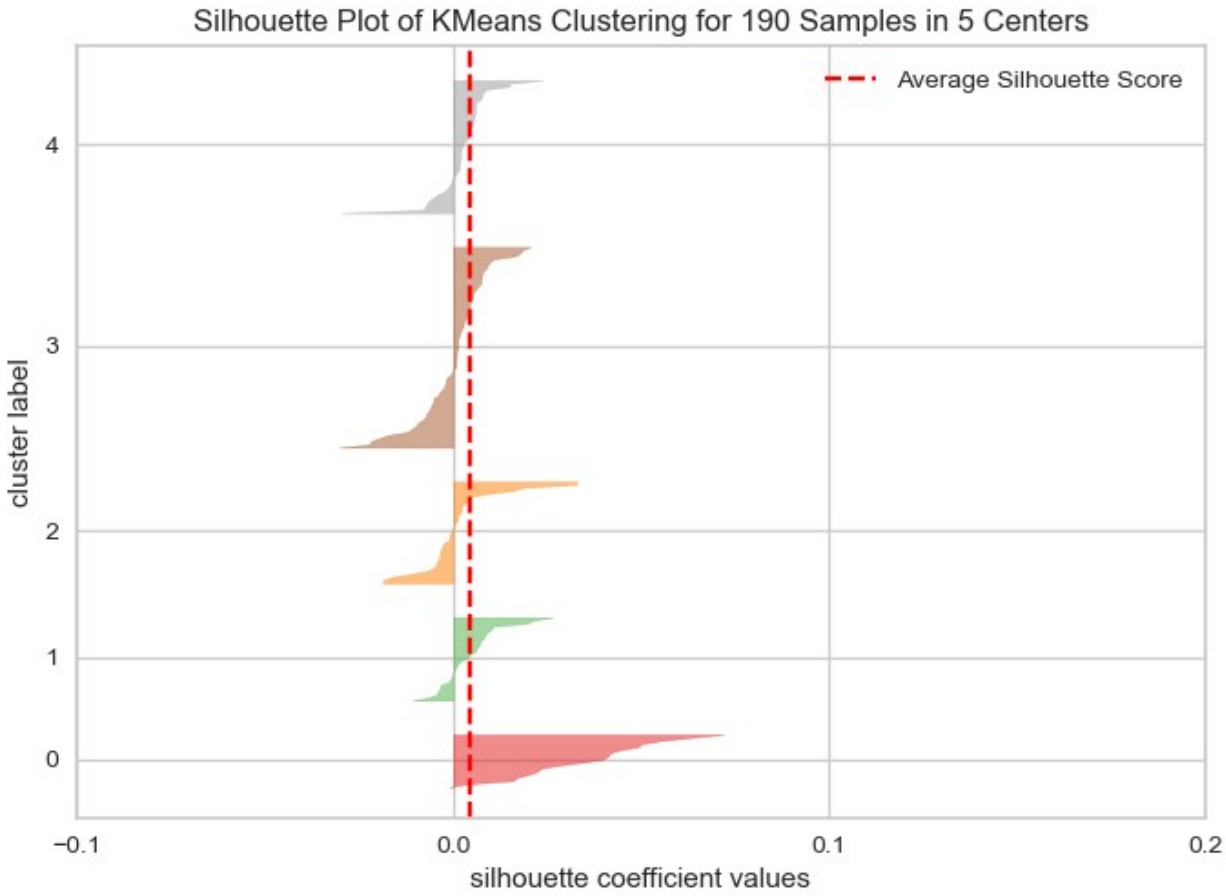


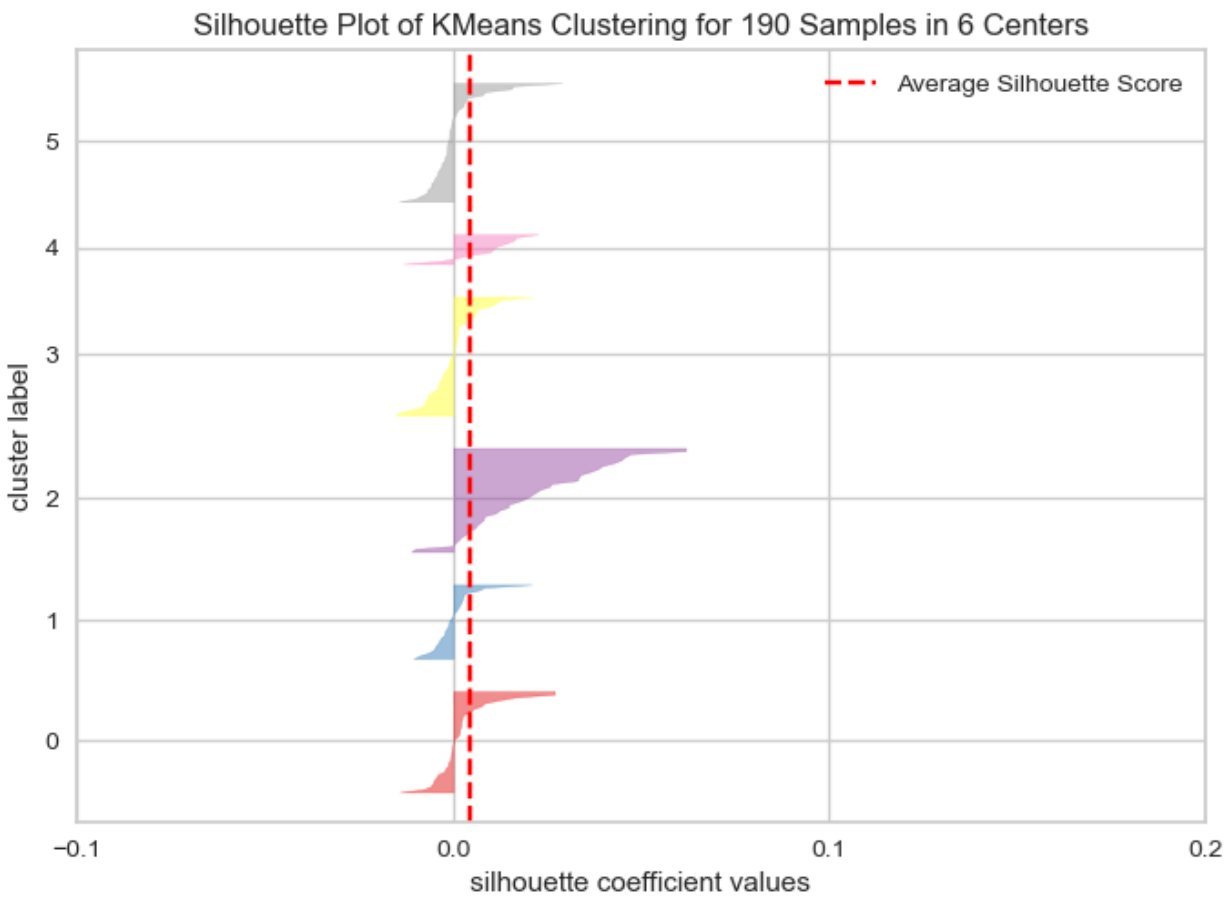

```python
# Appliquer la méthode silhouette pour différentes valeurs de k
for n_clusters in range(2, 11):
    model = KMeans(n_clusters=n_clusters)
    visualizer = SilhouetteVisualizer(model)

    visualizer.fit(X)        # Fit the data to the visualizer
    visualizer.show()        # Finalize and render the figure

# Afficher les graphiques de la méthode silhouette
plt.show()
```

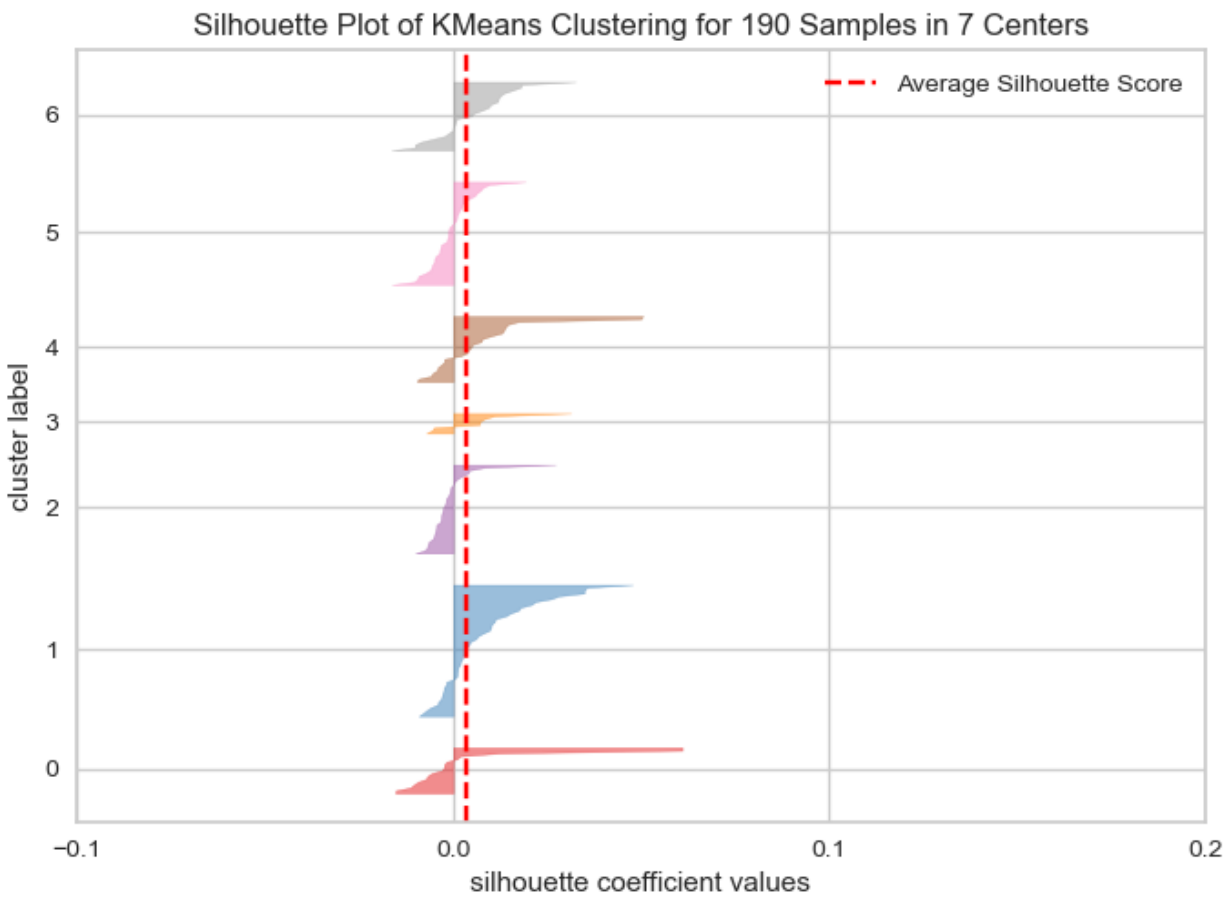
Silhouette Plot of KMeans Clustering for 190 Samples in 2 Centers
Average Silhouette Score
1
0
cluster label
−0.1
0.0
0.1
silhouette coefficient values

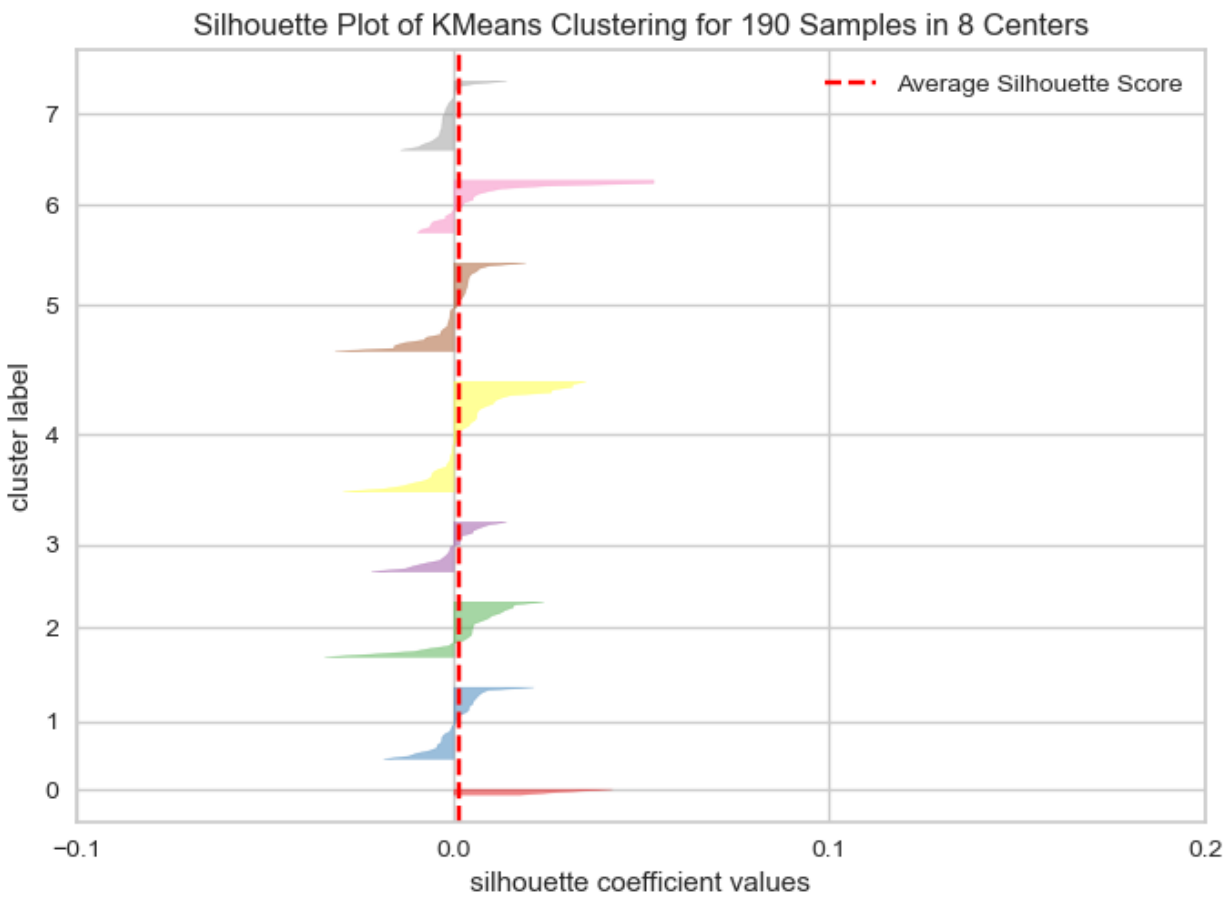
Silhouette Plot of KMeans Clustering for 190 Samples in 3 Centers
Average Silhouette Score
cluster label
2
1
0
−0.1
0.0
0.1
silhouette coefficient values

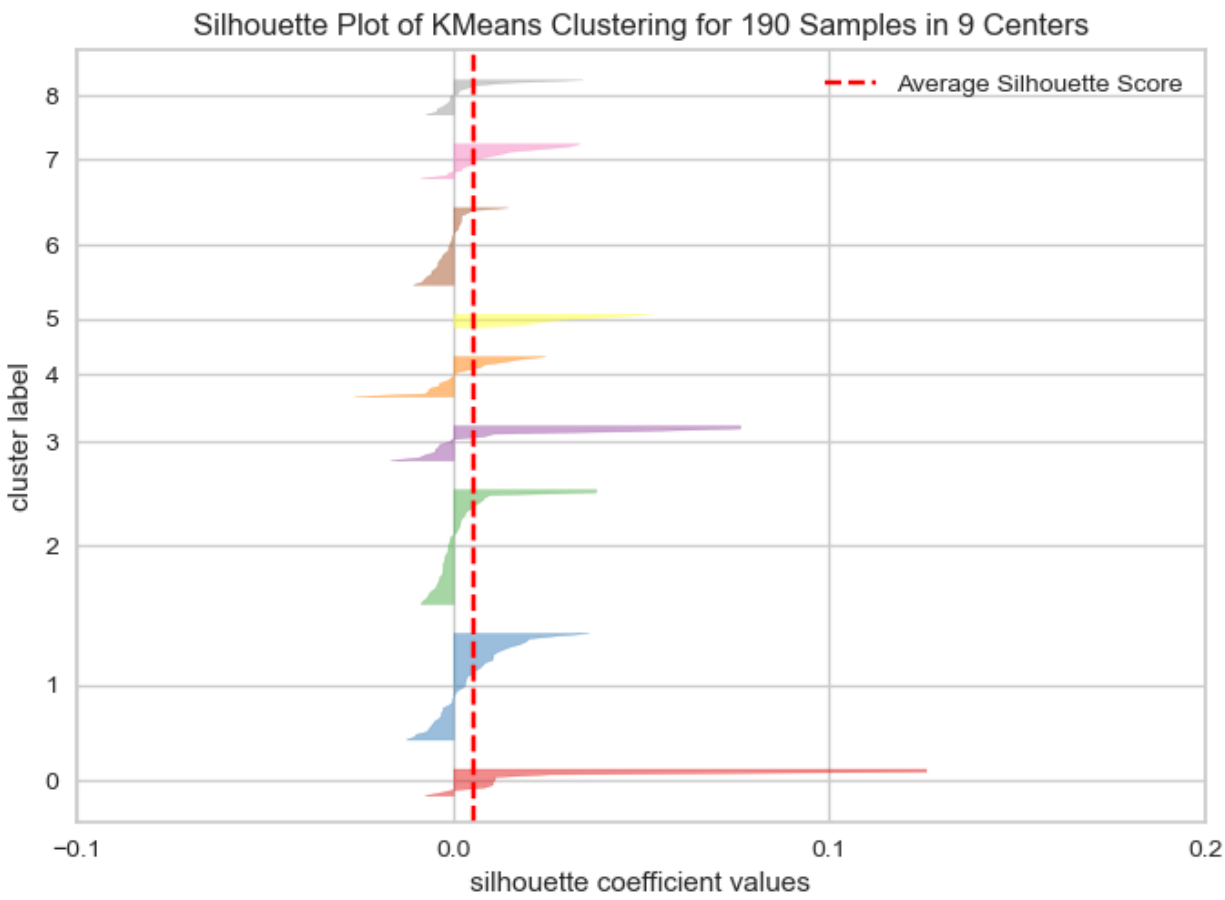

Silhouette Plot of KMeans Clustering for 190 Samples in 4 Centers
Average Silhouette Score
cluster label
3
2
1
0
−0.1
0.0
0.1
0.2
silhouette coefficient values

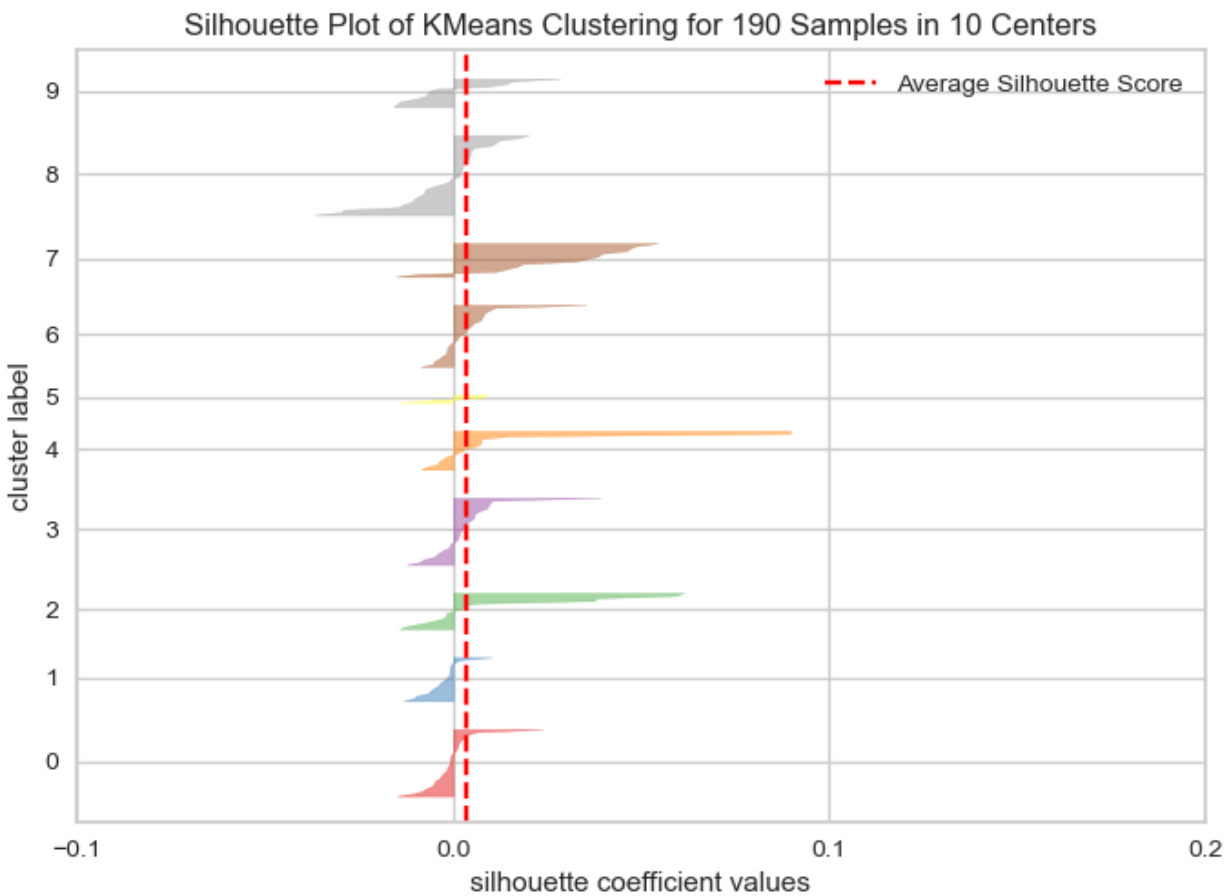

Silhouette Plot of KMeans Clustering for 190 Samples in 5 Centers
Average Silhouette Score
cluster label
4
3
2
1
0
−0.1
0.0
0.1
0.2
silhouette coefficient values


Silhouette Plot of KMeans Clustering for 190 Samples in 6 Centers
Average Silhouette Score
cluster label
5
4
3
2
1
0
−0.1
0.0
0.1
0.2
silhouette coefficient values


Silhouette Plot of KMeans Clustering for 190 Samples in 7 Centers
Average Silhouette Score
cluster label
6
5
4
3
2
1
0
−0.1
0.0
0.1
0.2
silhouette coefficient values


Silhouette Plot of KMeans Clustering for 190 Samples in 8 Centers
Average Silhouette Score
cluster label
7
6
5
4
3
2
1
0
−0.1
0.0
0.1
0.2
silhouette coefficient values


Silhouette Plot of KMeans Clustering for 190 Samples in 9 Centers
Average Silhouette Score
cluster label
8
7
6
5
4
3
2
1
0
−0.1
0.0
0.1
0.2
silhouette coefficient values



d. Ajouter au dataset original une variable “cluster”

```
import pandas as pd
import numpy as np
import re
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.cluster import KMeans
from sklearn.decomposition import TruncatedSVD
import matplotlib.pyplot as plt
import seaborn as sns

# Charger les données
df = pd.read_csv('./anthems.csv')

# Nettoyage superficiel
def clean_text(text):
    text = text.lower()
    text = re.sub(r'<.*?>', '', text)
    text = re.sub(r'[^a-zA-Z\s]', '', text)
    text = text.strip()
    return text
```

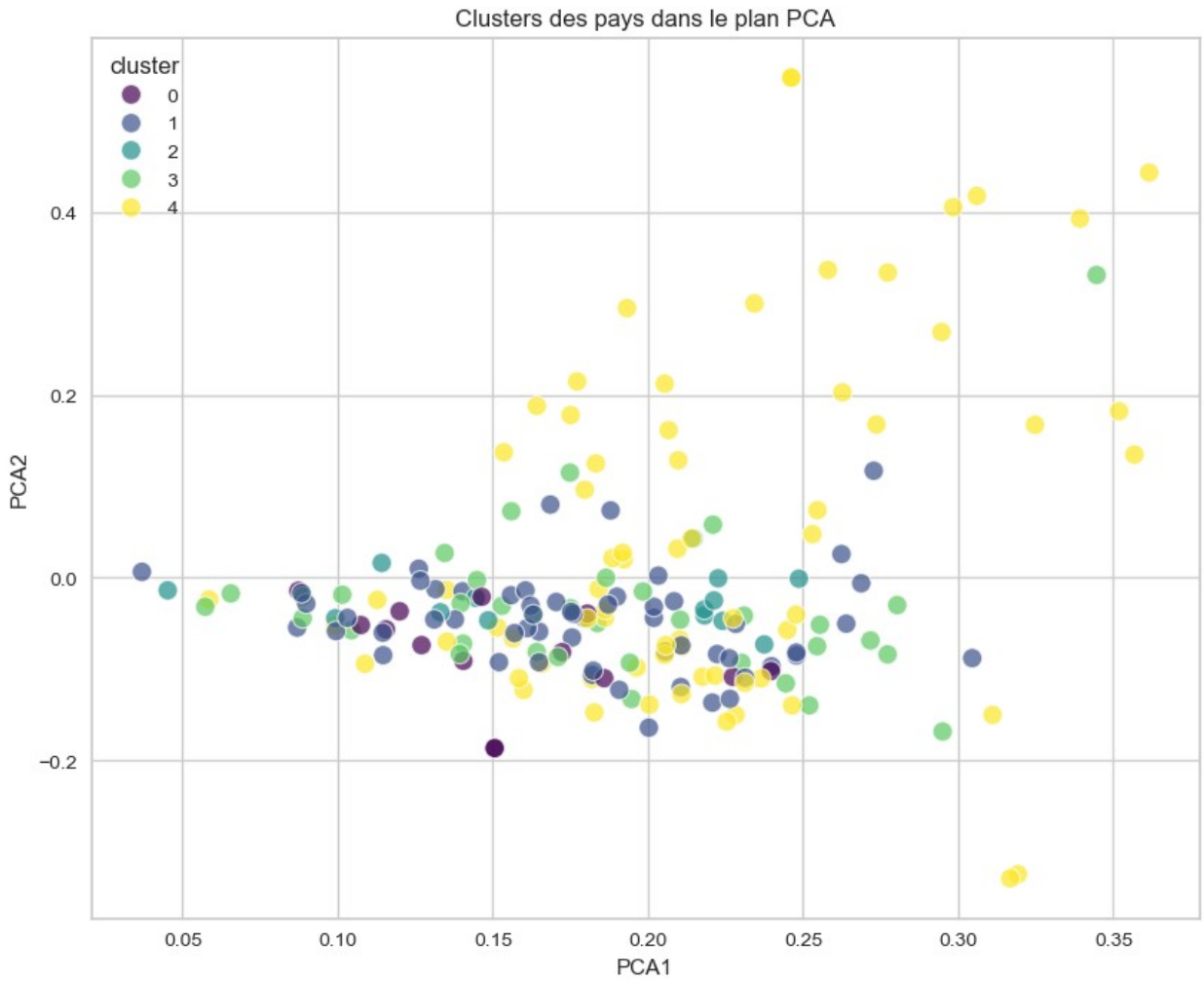
```python
df['Anthem'] = df['Anthem'].apply(clean_text)

# Vectorisation
vectorizer = TfidfVectorizer(stop_words='english')
X = vectorizer.fit_transform(df['Anthem'])

# Convertir la matrice sparse en une matrice dense
X_dense = X.toarray()

# Déterminer le nombre optimal de clusters (par exemple k=5)
optimal_k = 5
kmeans = KMeans(n_clusters=optimal_k, random_state=42)
clusters = kmeans.fit_predict(X_dense)

# Ajouter les clusters au dataset original
df['cluster'] = clusters

# Sauvegarder le dataframe avec les clusters
df.to_csv('./anthems_with_clusters.csv', index=False)

# Vérifier les premières lignes du dataframe avec les clusters
print(df.head())
```

```
      Country Alpha-2 Alpha-3 Continent  \
0     Albania      AL     ALB    Europe
1     Armenia      AM     ARM    Europe
2     Austria      AT     AUT    Europe
3  Azerbaijan      AZ     AZE    Europe
4     Belarus      BY     BLR    Europe

                                                Anthem  cluster
0  around our flag we stand united with one wish ...        1
1  our fatherland free independent that has for c...        4
2  land of mountains land by the river land of fi...        2
3  azerbaijan azerbaijan the glorious fatherland ...        4
4  we belarusians are peaceful people wholehearte...        3
```

e. Représenter les pays dans le plan engendré par les deux premières composantes principales

```python
# Réduction de dimension avec TruncatedSVD
svd = TruncatedSVD(n_components=2)
components = svd.fit_transform(X)

# Ajouter les composantes principales au dataframe
df['PCA1'] = components[:, 0]
df['PCA2'] = components[:, 1]

# Visualisation des clusters dans le plan PCA
plt.figure(figsize=(10, 8))
```

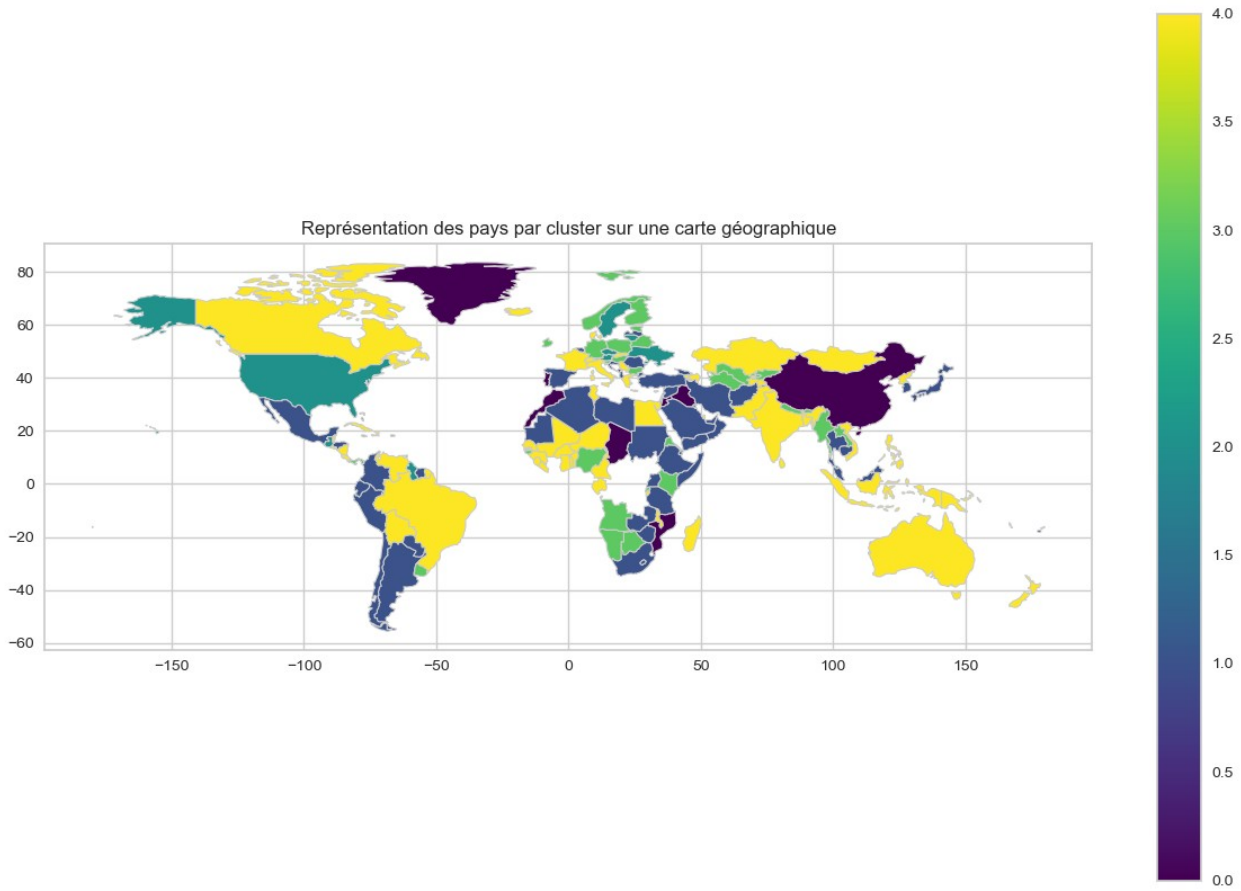
```
sns.scatterplot(x='PCA1', y='PCA2', hue='cluster', data=df,
palette='viridis', s=100, alpha=0.7)
plt.title('Clusters des pays dans le plan PCA')
plt.show()
```



f. Représenter les pays sur une carte géographique avec une couleur par cluster

```
import geopandas as gpd
import matplotlib.pyplot as plt

# Charger une carte du monde
world = gpd.read_file(gpd.datasets.get_path('naturalearth_lowres'))

# Fusionner avec le dataframe contenant les clusters
merged = world.set_index('name').join(df.set_index('Country'))

# Représenter les pays avec une couleur par cluster
```

```
fig, ax = plt.subplots(1, 1, figsize=(15, 10))
merged.plot(column='cluster', cmap='viridis', linewidth=0.8, ax=ax,
edgecolor='0.8', legend=True)
plt.title('Représentation des pays par cluster sur une carte
géographique')
plt.show()
```



g. Représenter les mots les plus fréquents dans chaque cluster avec un nuage de mots

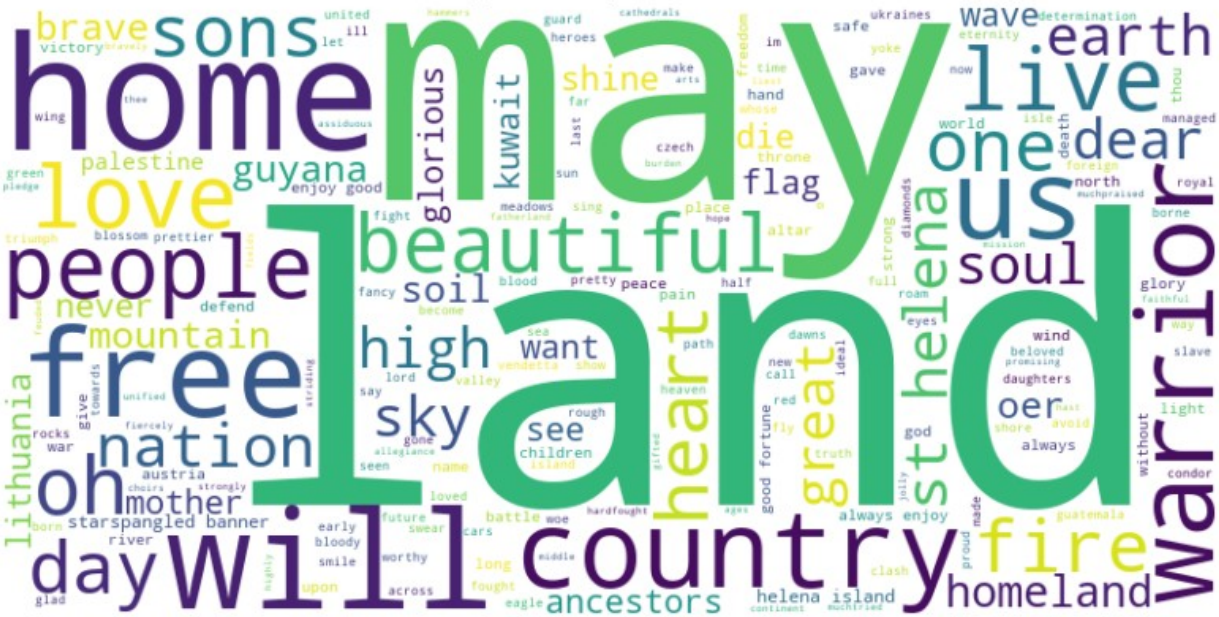
```
from wordcloud import WordCloud

# Créer des nuages de mots pour chaque cluster
for cluster in range(optimal_k):
    cluster_texts = df[df['cluster'] == cluster]
['Anthem'].str.cat(sep=' ')
    wordcloud = WordCloud(width=800, height=400,
background_color='white').generate(cluster_texts)

    plt.figure(figsize=(10, 5))
    plt.imshow(wordcloud, interpolation='bilinear')
    plt.axis('off')
```

```
plt.title(f'Nuage de mots pour le cluster {cluster}')
plt.show()
```

Nuage de mots pour le cluster 0

will
us
homeland
glory
people
country
march
path
king
god
see
live
flag
one
head
sea
arise
may
waving
new
every
want
children
life
brave
let
arm
lift
heart
fatherland
world
land
light
sun
time
goal
day
enemys
gunfire
invader
slaves
righto
symbol
victorious
honourable
honoured
proud
freedom
enemies
reaching
raise
call
flesh
glorious
faithful
high
portugal
banners
century
soil
establish
beloved
stars
lead
flutters
comeback
air
blood
immortal
mozambique
forever
eternal
miserable
eyes
oh
generation
victoriously
mother
valiant
victory
storied
unity
fate
old
china
liberty
heroes
pole
roaring
forefathers
roar
great
sky
conquered
undefeated
road
die
kisses
ocean
moves
love
rock
long
perished
chad
east
rises
defiance
beautiful
hills
salute
within
born
gave
must
dawn
now
sing
everybody
macedonian
arab
feet
haze
echo
today
hoist
voice
rays
hand
won
gunfire
march
rising
cause
protector
heights
forum
revolution
macedonia
refuse
europe
future
german
agent
feels
offensek
work
eminence
war
nation
splendor
millions
strong
work
mind
lion
independence
truth
sword
last
place
chest
son
make
know
field
spirit
korean
death
grant
great
march
bright
spain
soil
life
stand
roar
death
sing
bless
god bless
solemn
vow
country
sun
strong
zambia
voice
three
three centuries
homeland lifted
queen
soul
star
lord
victory
fiji
hope
arm
greatest
happiness
leader
new
blessed
national
wave
sacred
children
peace
proud
work
gave
law
honor
father
banner
tyrant
cry
love
great
king
sea
long
justice
chile
good
tear
sovereign
rise
chain
breast
promise
come
whose
soldier
south
heaven
fight
war
cause
high
eternal
give
faith
protect
defend
break
see
take
ancestor
song
resound
repeat
let
sacrifice
heard
fatherland
united
earth
joy
die
always
together
two
o
world
noble
hand
long live
symbol
sings
raise
struggle
glorious
salute
every
martyr
praise
light
unity
never
people
sky
blue
motherland
well
shore
blessing
immortal
tear
forehead
blessed
greatest
happiness
leader
national
wave
free
blood
voice
three
three centuries
zambia
god save
dear
upon
brother
battle
pride
thousand
now
call
today
keep
homeland
mountain
nation
salute
shine
symbol
songs
humiliated
around
future
hymn
wish
moist
great
korean
spirit
march
grant
death
brave
stand
roar
laurel
man
men
slave
gold
sound
know
field
strength
place
last
truth
sword
lion
independence
day
crown
heroes
name
glory
let us
light
son
africa
heart
reign
liberty
live
flag
freedom
forever
witness
make
country
oh
may
land
will
us
free
blood
people
sky
blue
joy
always
one
united
fatherland
eternal
rise
long
sea
king
great
cry
tyrant
love
proud
work
gave
law
honor
father
banner
peace
queen
soul
star
lord
hope
arm
fiji
victory
children
sacred
national
wave
blessing
immortal
tear
forehead
blessed
greatest
happiness
leader
new
homeland lifted
three
three centuries
zambia
voice
strong
sing
bless
death
solemn
vow
bright
soil
life
sun
country
god

Nuage de mots pour le cluster 2



Nuage de mots pour le cluster 3

Nuage de mots pour le cluster 4

land
thee
us
free
let
people
may
heart
one
will
love
god
country
canada
liberty
stand
glory
hail
flag
nation
slave
time
great
peace
die
thou
freedom
fatherland
honour
beloved
unity
homeland
arise
happiness
glorious
strong
son
hand
proud
light
rise
guard
ancestors
united
blood
future
forever
death
salute
soul
keep
always
mighty
voice
hear
haiti
lead
indonesia
hill
might
help
shout
path
mali
battle
name
thou art
join
banner
today
loyal
become
never
thousand years
dignity
beautiful
every
faith
men
unite
tear
malawi
brazil
noble
heaven
others
way
work
sound
mountain
star
world
children
without
we
oh
true
must
thine
canada o
arm
sublime
now
sea
island
song
swear
valley
toil
wave
well
praise
spirit
happy
together
command
soil
earth
home
cause
face
towards
field
fyrbnt
icelands
thousand
arms citizens
pledge
thrill
joy
gift
golden
war
fear
living
day
africa
ready
call
many
women
victory
make
mother
old
give
long live
young
spread
sun
strength
high
dear
fair
build
sky
man
eternal
brother
life
lord
bless
courage
good
night
truth
fight
raise
rich
sacred
destiny
brave
union
defend
god
papua new
native land
everywhere
father
hope
right
justice
last
blessed
firm
pray
progress
shout
path
help
indonesia
hill
might
strong
banner
today